

Sai Hemanth Kilaru

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EDUCATION

University of Arizona

Master of Science in Data Science

Tucson, AZ

May 2026

SRM Institute of Science and Technology

B.Tech in Computer Science and Engineering

Chennai, India

Jun 2024

EXPERIENCE

Research Data Support Specialist I

University of Arizona

Jul 2026 – Present

Tucson, AZ

- Architected tenant-isolated backend analytics with PostgreSQL row-level security and event-sourced replay, validated against 10,000-account/165,000-report and 40-account/600-report fixtures with zero leakage.
- Delivered privacy-preserving Admin and Health Official dashboards with small-cell suppression and account-level access control across backend, API, and frontend, passing all 42 frontend tests with zero console errors.

Machine Learning Intern

AICTE

Sep 2023 – Nov 2023

- Built Python, SQL, Pandas, and scikit-learn pipelines spanning preprocessing, feature engineering, model training, and statistical evaluation for supervised classification tasks across multiple datasets.
- Improved classification validation performance 10–15% through hyperparameter tuning and systematic error analysis, and shortened model iteration cycles 25% by automating experiment tracking and reporting.

Data Analytics Intern

AICTE

May 2023 – Jul 2023

- Built AWS EC2, S3, RDS, and IAM ingestion and analytics workflows in Python and SQL for relational storage, querying, and reporting across cloud-hosted operational datasets.
- Reduced manual data-handling effort 30% through workflow automation, while designing secure VPCs, security groups, and CloudFront distributions and evaluating access, scalability, and cost trade-offs.

PROJECTS

EDITH – Windows-Native AI Assistant | *Python, Qwen, llama.cpp* | [Link](#)

Aug 2026 – Present

- Designed a Windows-native local-first assistant with a custom golden 3D interface via Win32/ctypes and ModernGL, routing tasks between a local Qwen3-4B model (llama.cpp) and Codex with typed approvals and full audit trail.
- Integrated Faster Whisper speech recognition and Silero voice-activity detection with document understanding; verified 159 backend and 9 frontend tests passing with clean shutdown and zero orphaned tasks.

Community Surveillance Data Ingestion API | *Python, FastAPI, SQLite* | [Link](#)

Jun 2026 – Present

- Built a FastAPI/SQLite One Health microservice during a 5-day EpiHack, exposing seven endpoints with client/server bearer-token roles and pagination across three synthetic domains (Human, Animal, Environment).
- Designed hidden-reference filtering and Bedrock Claude ingestion with token logging, keeping every client read fully database-backed with zero LLM calls on the hot path.

LLM-Generated Social Networks | *Python, NetworkX, multi-model eval* | [Link](#)

Jan 2026 – May 2026

- Led a five-person research team building a reproducible pipeline evaluating three GPT-4.1 variants across four cultures, languages, and generation methods on a fixed 50-persona roster; accepted at NeurIPS 2026.
- Passed 72/72 method-study and 96/96 language-study validation checks, isolating true model-behavior differences and quantifying edge distance: 0.074-0.081 vs. 0.119-0.126 for the smaller model.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript

Frameworks: FastAPI, Flask, React, LangChain, Streamlit, Django, LiteLLM

Developer Tools: PostgreSQL, Git, GitHub, Docker, Pytest, CI, REST APIs, AWS (EC2, S3, RDS, IAM)

Libraries: Pandas, NumPy, scikit-learn, NetworkX, Hugging Face Transformers, TensorFlow, OpenCV